OpenCV Stereo Depth Estimation Experiment Report

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1 Objective

This experiment uses the `aloeL.jpg` and `aloeR.jpg` images from the official OpenCV sample data to implement a complete pipeline from a stereo image pair to a disparity map and a relative-depth map. The focus is not on reimplementing a stereo-matching algorithm from scratch, but on connecting input validation, preprocessing, matching, numerical filtering, result saving, and evidence checking into a reproducible experimental loop.

1.1 Course relevance

Stereo depth estimation connects the geometric models of computer vision with practical image-processing interfaces. Disparity is the directly observable pixel displacement between two images, whereas depth is also affected by the focal length, baseline, and calibration quality. This experiment distinguishes between producing an image with visually layered structure and obtaining depth with physical units that can be used for measurement. It also illustrates how invalid matches, occlusions, and texture conditions affect the result.

1.2 Experimental objectives

1. Understand the pinhole stereo model, the epipolar constraint, and the relationship between disparity and depth;
2. Learn the OpenCV interfaces for image loading, grayscale preprocessing, and `StereoSGBM` stereo matching;
3. Convert the fixed-point disparity returned by the matcher into floating-point disparity in pixel units, and perform valid-region filtering and visualization;
4. Explain how parameters such as `minDisparity`, `numDisparities`, the smoothness penalties, and speckle filtering affect the matching result;
5. When camera calibration parameters paired with the image pair are unavailable, compute and analyze a relative-depth proxy without physical units;
6. Save the left and right images, disparity map, valid-disparity mask, arrays, and parameter summary generated by the real run, and use them to verify the results.

1.3 Reproducibility and input/output scope

This report uses the aloe stereo image pair from OpenCV's official `4.x/samples/data` directory as the only primary experimental input. The program command, software versions, matching parameters, output-array dimensions, and statistics are taken from `task2/results/params.`

`json` and `run_evidence.md` generated by this run. The outputs include pixel disparity and depth_proxy $= 1/d$ on an arbitrary scale. Because the current image pair has no verified focal length, baseline, or **Q** matrix, this proxy is not interpreted as metric depth in metres, centimetres, or any other physical unit.

2 Principle

2.1 Pinhole stereo model

Let the optical centres of the left and right cameras be separated horizontally by the baseline B, and let the two cameras have the same focal length f. For a spatial point (X, Y, Z), its horizontal projections under the parallel-camera model can be written as

$$x_L = \frac{fX}{Z}, \qquad x_R = \frac{f(X-B)}{Z}. \tag{1}$$

Subtracting the two equations gives the relationship between disparity and depth:

$$d = x_L - x_R = \frac{fB}{Z}, \qquad Z = \frac{fB}{d}. \tag{2}$$

Thus, a larger disparity generally corresponds to a closer scene point, whereas a smaller disparity corresponds to a farther point. Equation (2) also shows that pixel disparity alone cannot determine the physical depth scale: the focal length and baseline associated with the current camera system are also required.

2.2 Epipolar constraint and horizontal search

Corresponding points in a stereo image pair satisfy the epipolar constraint defined by the fundamental matrix:

$$\mathbf{x}_R^{\mathsf{T}} \mathbf{F} \mathbf{x}_L = 0, \tag{3}$$

where $\mathbf{x}_L$ and $\mathbf{x}_R$ are the corresponding homogeneous pixel coordinates and $\mathbf{F}$ is the fundamental matrix. This relationship restricts the candidate point in the right image to the epipolar line associated with a point in the left image. After reliable stereo rectification, the epipolar lines are approximately horizontal scanlines, so the matcher only needs to search for horizontal displacements within the same row.

This experiment follows the direct matching procedure for the official OpenCV aloe sample and does not re-estimate a rectification transform. The disparity convention in this report is $d = x_L - x_R$; the valid-disparity rule used by the program is a finite value with $d > 15$.

2.3 StereoBM and StereoSGBM

StereoBM compares image blocks within a local window. It is structurally simple and fast, but it is prone to holes or incorrect matches in textureless regions, at occlusion boundaries, and in repeated textures. **StereoSGBM** accumulates matching costs along multiple directions and uses semi-global path optimization to balance local consistency against global smoothness. It generally produces more continuous disparity results. This experiment selects StereoSGBM so that the official aloe images yield an interpretable course demonstration.

For a pixel p and candidate disparity d, the matching cost can be abstracted as $C(p, d)$. The recurrence along a path penalizes changes in the disparity of adjacent pixels: a small change is assigned the penalty P_1, whereas a larger jump is assigned the penalty P_2. This experiment uses $P_2 > P_1$, encouraging most regions to remain smooth while allowing disparity discontinuities at object boundaries. The actual configuration also includes the following constraints:

- `minDisparity` and `numDisparities` define the search origin and range; their values in this run are 16 and 96, respectively;
- `blockSize` determines the local matching window and is 5 in this run; a larger window is generally more stable but loses boundary detail;
- `uniquenessRatio` requires the best match to be clearly better than the second-best match, reducing the probability that ambiguous points are retained;
- `disp12MaxDiff` is used for the left–right consistency check;
- `speckleWindowSize` and `speckleRange` remove small regions with anomalous disparity changes.

2.4 Relative depth and calibration boundary

The OpenCV matcher returns an integer result with a fixed scale. The program divides it by 16 to obtain floating-point disparity in pixel units. For a valid disparity, it computes

$$\text{depth_proxy} = \frac{1}{d}, \qquad d > 15. \tag{4}$$

This quantity preserves the front-to-back ordering represented by disparity, but it does not contain the factor fB in Equation (2). The current official aloe sample does not provide an explicitly paired f, B, `doffs`, or $\mathbf{Q}$. Therefore, the experiment does not call `reprojectImageTo3D` with guessed parameters and does not generate a point cloud with physical scale. If `aloeGT.png` is used, it can only serve as a disparity reference image; it cannot be treated as metric-depth ground truth.

2.5 Depth sensitivity to disparity error

Equation (2) gives not only the depth inversion formula but also the direction of error propagation. Differentiating with respect to d for fixed fB gives

$$\frac{\partial Z}{\partial d} = -\frac{fB}{d^2}, \qquad \frac{\Delta Z}{Z} \approx -\frac{\Delta d}{d}. \tag{5}$$

For the relative-depth proxy used in this report, we likewise have

$$\frac{\partial (1/d)}{\partial d} = -\frac{1}{d^2}. \tag{6}$$

This means that a disparity perturbation of the same magnitude has a larger relative effect in a low-disparity region, whereas in a high-disparity region, which is generally closer to the camera, the same pixel error produces a smaller absolute change in $1/d$. Consequently, the grayscale continuity of a disparity map alone is insufficient to judge depth precision at every pixel; the valid-disparity mask, local texture, and matching consistency must also be considered.

2.6 Definition and statistical meaning of valid disparity

The program defines a valid disparity as a finite pixel satisfying $d > 15$ and stores all other positions as `NaN`. This threshold is first a numerical and search-range constraint: the matcher starts its search at 16, so values below the threshold cannot be directly treated as reliable positive disparity. The rule answers which pixels enter the subsequent computation; it is not equivalent to saying that all retained pixels are correctly matched. Thus, the valid-disparity

mask is the support set for later statistics rather than an accuracy label, and the valid-pixel fraction is coverage rather than accuracy.

When reliable disparity ground truth is unavailable, at least three kinds of information should be recorded separately: coverage, namely how many pixels pass the validity filter; the disparity distribution, namely the range and quantiles of the retained pixels; and matching confidence, namely whether the disparities pass left–right consistency, reprojection-error, or external-reference checks. This experiment saves the raw `npy` arrays and the valid-disparity mask precisely to avoid treating a percentile-stretched display PNG as quantitative evidence of any of these three quantities.

2.7 Coordinate comparability after downsampling

In this run, the left and right images are synchronously downsampled once using `pyrDown`. The matching coordinates therefore lie on the 641×555 processed images rather than on the original 1282×1110 images. One pixel in the disparity array corresponds to one pixel in the processed-image coordinate system. If the result is compared with calibration parameters defined at the original resolution, or with another result that was not downsampled, the scale transformations of the focal length, principal point, and disparity coordinates must be considered consistently. Otherwise, a numerical difference caused solely by resolution changes may be misinterpreted as a depth change even when the scene and algorithm are identical.

The immediate benefit of downsampling is reduced matching cost, allowing the course experiment to be reproduced quickly; the cost is that fine textures and narrow boundaries may be smoothed or merged. Because this experiment interprets d and $1/d$ only within the same processed resolution, the report does not extrapolate the result to metric depth in the original camera coordinate system. If calibration parameters are enabled in future work, the calibration model should be regenerated at the same resolution, or the parameters should be explicitly converted to the processed resolution.

3 Experimental Environment and Data

3.1 Official data source

The experiment uses `aloeL.jpg` and `aloeR.jpg` from the `4.x/samples/data` directory of the official OpenCV repository, `opencv/opencv`. This image pair was selected because it is an official OpenCV sample with an explicit left–right relationship, and the official Python stereo-matching example also provides a reproducible SGBM processing strategy for the aloe images. It therefore supports the course experiment without introducing an additional dataset.

Both original JPG images have dimensions of 1282×1110 and contain three channels. The program synchronously applies `pyrDown` once to the left and right images, giving an actual matching size of 641×555. `aloeGT.png` in the same official directory is an optional disparity reference image; this experiment does not treat it as a calibration parameter, a depth scale, or an error ground truth.

Table 1: Experimental data and software environment

Item	Measured version or description
Data directory	Official OpenCV `4.x/samples/data`
Input files	`aloeL.jpg`, `aloeR.jpg`
Original dimensions	1282×1110, three channels
Python	3.13.5
OpenCV runtime	5.0.0
NumPy	2.5.3

Version information, input paths, and run parameters come from the program-generated `task2/results/params.json`. The SHA-256 values and download URLs of the input files are recorded in `task2/results/run_evidence.md`, allowing the experiment to be checked in the same environment without relying on image statistics copied manually into the report.

3.2 Actual run command

The results in this report were generated by the following default command. The run did not use `--no-pyrdown` and did not apply any additional parameter adjustment:

```
task2/.venv/bin/python task2/src/stereo_depth.py \\
  --left task2/data/aloeL.jpg \\
  --right task2/data/aloeR.jpg \\
  --out-dir task2/results
```

The program first checks that both images can be read and that their dimensions agree. It then performs grayscale conversion, synchronized downsampling, StereoSGBM matching, disparity filtering, and relative-depth computation. No calibration file from another image pair is used; all results come from the outputs in the root directory corresponding to the command above.

4 Program Design and Implementation

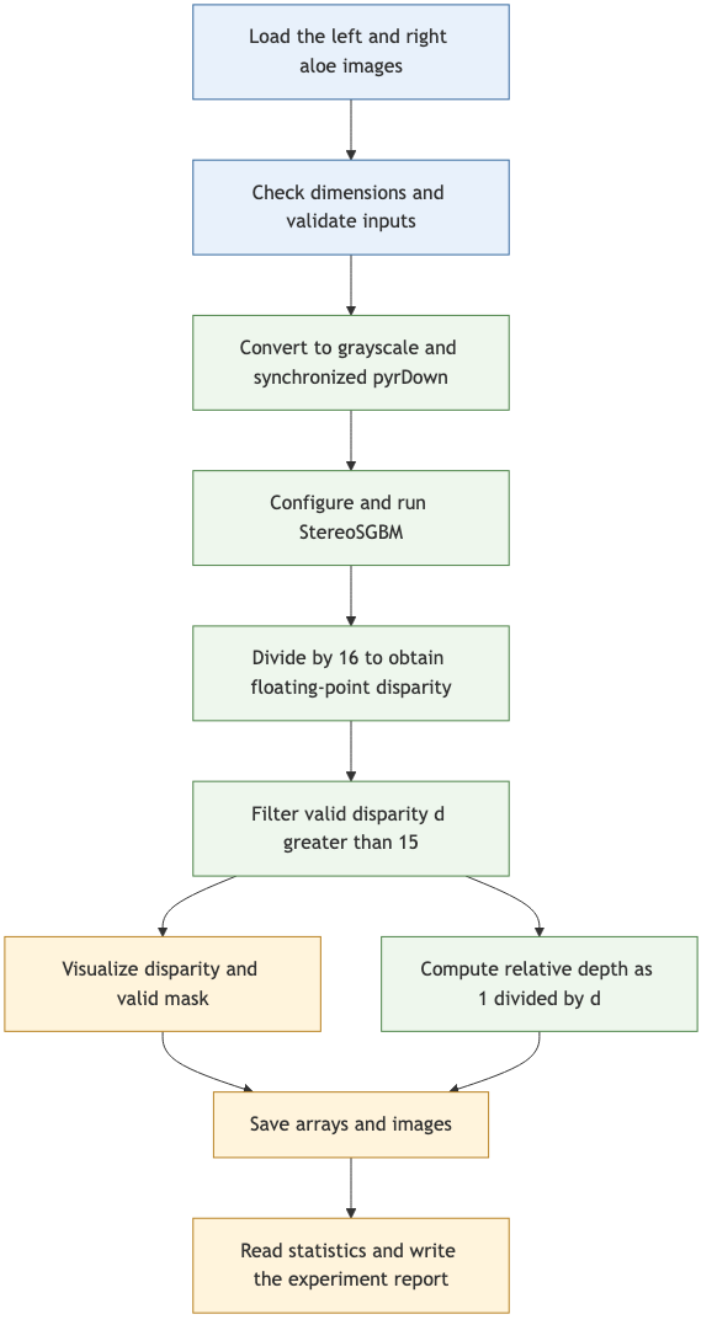


Figure 1: Stereo depth estimation pipeline generated from the Mermaid source file.

The program entry point is `task2/src/stereo_depth.py`, which implements a single reproducible run from input validation to result serialization. The program does not use display images as numerical inputs; instead, it saves both the original floating-point arrays and the 8-bit visualizations used in the report. The overall processing relationship is shown in Fig. 1.

4.1 Input Reading, Validation, and Preprocessing

The program uses `cv2.imread` to read the left and right color images. Before matching, it checks that the files exist, that OpenCV returns non-empty arrays, and that the two images have the same row and column dimensions. If either image is empty or the dimensions are inconsistent, the program raises a locatable error and does not continue to produce seemingly valid results.

The two images are then converted to grayscale. To reduce the computational cost, the left and right color images and their grayscale versions are downsampled once with `cv2.pyrDown`. Thus, no different scaling ratio is introduced between the two views. The original size of 1282×1110 becomes 641×555; all subsequent disparity arrays and output images correspond to this processed size.

4.2 StereoSGBM Configuration

The parameters used in the actual run are recorded in `task2/results/params.json` and listed in Table 2. `numDisparities` is set to a multiple of 16 so that the search range covers the main disparity variation in the official aloe images.

Table 2: StereoSGBM parameters used in the reported run

Parameter	Actual value
`window_size`	3
`minDisparity`	16
`numDisparities`	96
`blockSize`	5
P_1	72
P_2	288
`disp12MaxDiff`	1
`uniquenessRatio`	10
`speckleWindowSize`	100
`speckleRange`	32

4.3 Disparity, Valid-Disparity Mask, and Relative Depth

The OpenCV matcher returns integer disparities with a fixed scale factor. The program divides these values by 16 to obtain a floating-point array in pixel units. In this run, a disparity is considered valid when it is finite and satisfies $d > 15$; all other positions are stored as `NaN` in `disparity_raw.npy`, so they do not enter the range, proportion, or relative-depth statistics. The core logic is:

```
disparity = matcher.compute(left_match, right_match).astype(np.float32) / 16.0
valid = np.isfinite(disparity) & (disparity > 15)
disparity_raw = np.where(valid, disparity, np.nan)
depth_proxy = np.where(valid, 1.0 / disparity_raw, np.nan)
```

The relative-depth array is computed as $1/d$ and has no physical unit. It is used only to express relative front-to-back ordering. To generate a stable display image, the program takes the first and 99th percentiles of the valid pixels as the display range and linearly maps that range to 0–255. Therefore, the grayscale values in the PNG files are not the raw disparities or raw relative-depth values; quantitative analysis must read the `npy` arrays instead.

4.4 Output and Exception Checks

After writing each file, the program checks that the file exists and is non-empty. The outputs include `left.png`, `right.png`, `disparity_raw.npy`, `disparity.png`, `valid_mask.png`, `depth_proxy.npy`, `depth_proxy.png`, and `params.json`. The parameter summary also records the input paths, processed dimensions, matcher parameters, Python/OpenCV/NumPy versions, and output paths, so that the numerical values reported here can be traced back to the same set of result files.

5 Experimental Results

5.1 Generated Images and Data Reading

Figure 2 shows the left and right input images actually used by the program. Figure 3 presents, from left to right, the normalized disparity image, the valid-disparity mask, and the relative-depth visualization. All images are direct references to program outputs under `task2/results/`. The images are used to inspect spatial structure; quantitative conclusions are computed from `disparity_raw.npy` and `depth_proxy.npy`.



Figure 2: Official OpenCV aloe stereo input images (left: `aloeL.jpg`; right: `aloeR.jpg`).



Figure 3: Program-generated result images, from left to right: normalized disparity, valid-disparity mask, and arbitrary-scale relative depth.

The normalized PNG files use percentile-based ranges to enhance display contrast, so grayscale values in a PNG must not be substituted directly into $Z = fB/d$. For example,

the raw disparity at a pixel must be read from `disparity_raw.npy`; its relative-depth value must be read from `depth_proxy.npy`. Both arrays retain the `NaN` semantics of invalid regions.

5.2 Results from the Actual Run

The statistics below are read directly from the floating-point arrays generated by the program. The arrays have size 555×641, for a total of 355755 pixels. Among the finite disparity pixels satisfying $d > 15$, 261378 are valid, corresponding to 73.4713% of all pixels. The actual run results are summarized in Table 3.

Table 3: Measured results from the official OpenCV aloe sample

Metric	Measured value
Original image size	1282×1110
Matching image size	641×555
Floating-point array size	555×641
Total pixels	355755
Valid disparity pixels	261378
Invalid-pixel ratio	26.5287%
Valid-disparity ratio	73.4713%
Valid disparity range	16.000–105.875 px
Relative-depth range	0.0094451–0.0625 (arbitrary scale)
Formula verification error $\max \lvert z - 1/d \rvert$	0.0

5.3 Spatial Distribution of Valid-Matching Coverage

The global valid ratio only answers what fraction of all processed pixels passes the validity criterion; it does not indicate whether those pixels are distributed uniformly across the image. To examine this issue, the 641×555 matching image is divided into a 15×20 grid, and each cell is assigned the proportion of pixels that are finite and satisfy $d > 15$. The color range in Fig. 4 is fixed to 0–100%, allowing direct comparison between locations. Blank or light-colored cells indicate that more pixels at those locations failed the criterion; they do not indicate zero depth.

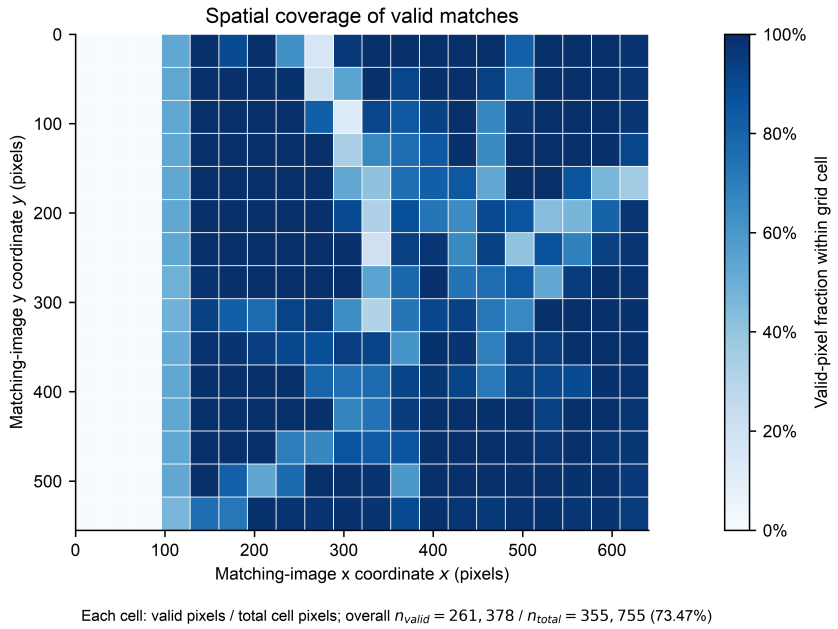


Figure 4: Spatial valid-matching coverage. Color denotes the proportion of pixels with finite disparity and $d > 15$ px in each cell of the 15×20 grid. The calculation covers all 261378 valid pixels, giving an overall coverage of 73.4713%.

Figure 4 shows that valid matches are not a uniform region that can be described only by one global percentage. The coverage varies substantially between grid cells, indicating that occlusion, insufficient texture, repeated texture, or search-boundary effects may be spatially concentrated. This figure is therefore a spatial-support diagnostic for the matching result: it helps locate regions for further inspection, but it is not a disparity-error map and its color values must not be interpreted as algorithmic accuracy.

5.4 Numerical Distribution of Valid Disparities

The 261378 valid disparity pixels have a range of 16.0000–105.8750 px and a median of 32.0000 px. The values of P_{05}, P_{25}, P_{75}, and P_{95} are 23.7500, 26.1875, 53.7500, and 65.7500 px, respectively. Figure 5 shows the corresponding histogram; the shaded region denotes the interquartile range. The valid pixels are not concentrated at a single disparity value. Instead, the matching displacement spans a relatively broad numerical range, which is consistent with surfaces at different front-to-back positions in the input scene.

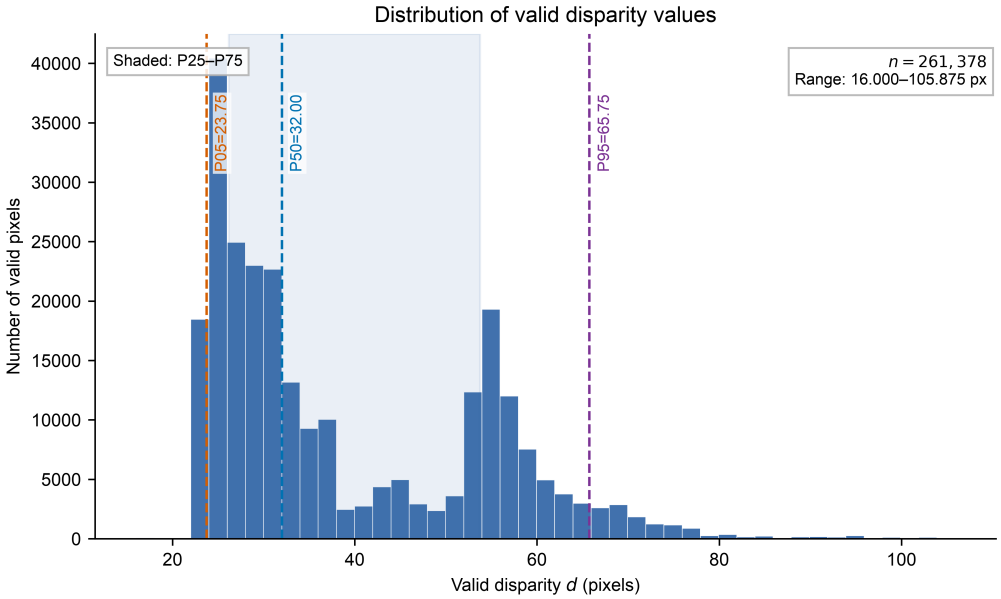


Figure 5: Distribution of valid disparities ($n = 261378$). Statistics use only finite pixels with $d > 15$ px; the shaded region denotes P_{25}–P_{75}, and dashed lines mark P_{05}, the median, and P_{95}. The horizontal axis is measured in pixels.

It is important to emphasize that Fig. 5 describes the numerical distribution produced by one StereoSGBM run; it is neither an error distribution nor a confidence interval. Under $z = 1/d$, a larger disparity corresponds to a smaller relative-depth proxy and can therefore be used to infer relative front-to-back ordering. However, without verified values of f, B, and $\mathbf{Q}$, these values cannot be converted to metric depth in meters and the width of the distribution cannot be interpreted as matching accuracy.

5.5 Spatial Disparity Heterogeneity and Quadrant Comparison

The global histogram still cannot answer where the larger or smaller disparities occur. Therefore, Fig. 6(a) displays the median valid disparity in the same 15×20 grid, while Fig. 6(b) divides the image at its center into upper-left, upper-right, lower-left, and lower-right quadrants and plots a box plot of all valid disparities in each region. The corresponding sample counts, regional coverage, medians, and interquartile ranges are listed in Table 4.

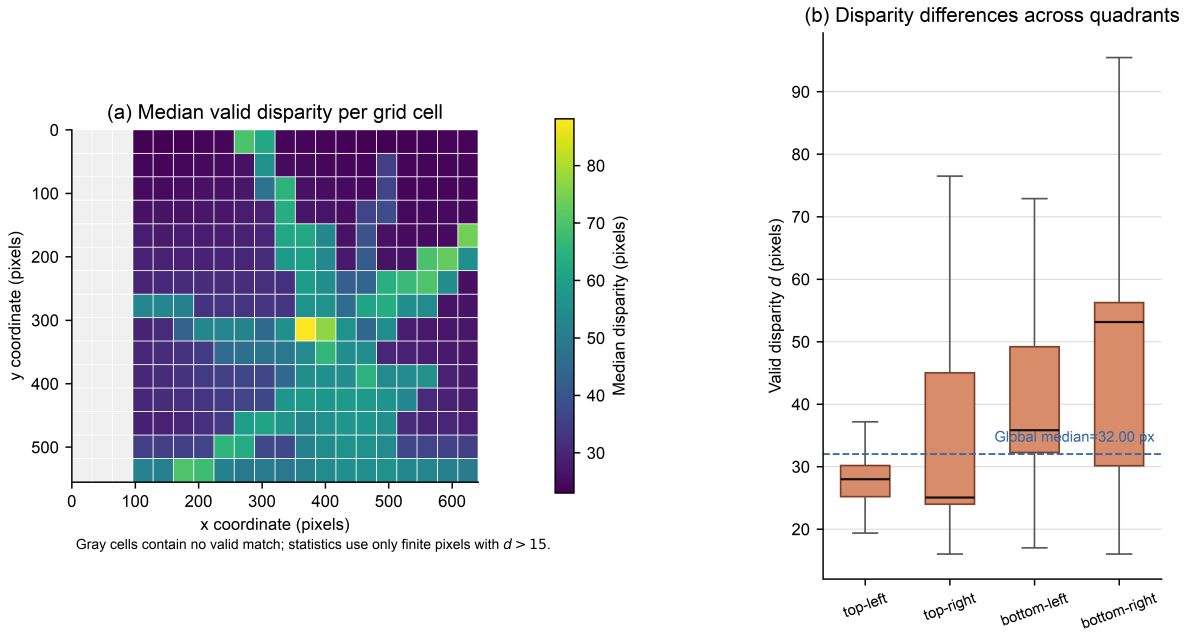


Figure 6: Spatial disparity and quadrant comparison. (a) Color denotes the median valid disparity within each grid cell; gray denotes no valid match. (b) Box plots of all valid disparities in the four quadrants; the box spans P_{25}–P_{75} and the horizontal line marks the median. The quadrants are defined only by image coordinates and do not represent semantic object classes.

Table 4: Valid-disparity statistics for the four image quadrants

Region	Valid samples	Coverage	Median disparity / px	P_{25}–P_{75} / px
Upper-left	51934	58.59%	28.0000	25.1875–30.1875
Upper-right	74778	84.10%	25.0625	24.0000–45.0000
Lower-left	52589	59.12%	35.8125	32.2500–49.1875
Lower-right	82077	91.98%	53.1250	30.1250–56.2500

The four regions differ simultaneously in coverage and median disparity. The lower-right region has the largest number of valid samples and the highest coverage, with a median disparity of 53.1250 px; the upper-right region has a substantially smaller median of 25.0625 px. Given the monotonic relationship $z = 1/d$, the reported output exhibits observable relative front-to-back variation across image locations. Because the quadrants are geometric partitions rather than object segments, these differences cannot be attributed directly to a particular object, nor can they be used to claim statistical significance or algorithmic superiority.

Figure 6 also shows within-region dispersion. For example, the boxes and whiskers in the upper-right and lower-right quadrants span broad ranges, indicating that a single quadrant contains pixels with different disparities at the same time. Thus, local depth interpretation should consider spatial position, sample coverage, and the numerical distribution together; a single high-disparity pixel should not be selected as the "nearest point."

5.6 Quantitative Interpretation of Relative Near–Far Ordering

After sorting the valid disparities, the lower 0–5% interval spans 16.000–23.750 px and has a median relative-depth proxy of 0.043127. The upper 95–100% interval spans 65.750–105.875 px and has a median relative-depth proxy of 0.014172. The latter is smaller, as required by the monotonic relationship $1/d$: larger disparity corresponds to a relatively nearer region.

The disparity and relative-depth images retain the main structures of the leaves, flowerpot, and background, indicating that StereoSGBM with fixed parameters has produced usable matches. Blank regions in the valid-disparity mask are not "zero depth"; they are invalid matches excluded from the statistics by the program.

5.7 Result Boundaries

Invalid regions may arise from occlusion, insufficient texture, repeated texture, or insufficient search range near the left and right image boundaries. They should not be filled with arbitrary values merely to make the visualization appear complete. If `aloeGT.png` is used, it can only serve as an official disparity reference image; it is not camera calibration or metric-depth ground truth. Because the current aloe stereo pair has no verified f, B, or $\mathbf{Q}$, this experiment cannot report absolute depth in meters or centimeters, nor can it provide a metric-scale 3D point cloud.

6 Discussion and Error Analysis

6.1 Main sources of error

Occlusion. A spatial point may be visible in only one of the left and right images. At an occlusion boundary, the matcher cannot find the true correspondence, which can produce holes, isolated disparities, or jumps near the boundary. Blank regions in the valid-disparity mask should be retained rather than being turned into fabricated measurements by interpolation.

Insufficient texture. In uniform or low-contrast regions, the matching costs of multiple candidate disparities are similar. `uniquenessRatio` therefore rejects results that are not sufficiently unambiguous. Large weak-texture regions in the tabletop and background are consequently more likely to contain invalid or unstable disparities.

Repeated texture. Periodic texture can produce several local matches with nearly identical costs. SGBM path smoothing can use neighbourhood structure to reduce some of this ambiguity, but it cannot guarantee the correct global correspondence on a repeated pattern.

Image boundaries and search range. The matching window and disparity search range consume pixels available near the left and right image boundaries. When the true disparity lies outside the search interval from `minDisparity` to `minDisparity+numDisparities`, no valid result can be obtained even if the image contains texture.

6.2 Effects of parameters and downsampling

This run records one fixed parameter setting and does not include an additional comparison experiment. In general, increasing `blockSize` can improve local stability in weak-texture regions, but it makes object boundaries thicker. A smaller window preserves more detail but is more sensitive to noise. `P1` and `P2` control disparity smoothness: values that are too large may flatten real boundaries, whereas values that are too small may produce speckles. Increasing `uniquenessRatio` leaves fewer but more reliable matches; the speckle parameters remove small connected regions with abnormal disparity.

Applying `pyrDown` once synchronously reduces the number of matching pixels and the computational burden, which is suitable for rapid reproduction in a course experiment. The cost is that fine leaf boundaries and small textures may be merged, reducing disparity detail. If detail is the priority, downsampling should be disabled while maintaining identical scaling for the left and right images, and a complete new set of arrays, images, and parameters should be recorded. Results at different resolutions must not be mixed in one table.

6.3 Limitation of the metric-depth scale

The current result measures pixel disparity only. According to $Z = fB/d$, even a perfectly estimated disparity still has an unknown proportional factor when fB is unknown. Therefore, the numerical value of `depth_proxy` cannot be interpreted as metres or centimetres. Other calibration files in the official sample directory have not been verified to match the aloe images, the resolution, and the camera setup, and therefore cannot be applied directly.

6.4 The valid-pixel fraction is not matching accuracy

The valid-pixel fraction of 73.4713% obtained in this run means, strictly, the fraction of pixels at the processed resolution that passed the filter "finite and $d > 15$". It reflects coverage, but it does not establish whether those disparities agree with the true correspondences. A pixel with a finite disparity may still result from repeated texture, an occlusion boundary, or an incorrect local cost minimum. Conversely, an invalid pixel may simply indicate that the matcher could not provide a sufficiently confident answer; it does not mean that the scene has no depth there. This experiment therefore discusses coverage, disparity distribution, and matching correctness separately, without describing the valid-pixel fraction as accuracy or recognition rate.

To obtain a comparable accuracy metric, an additional disparity reference with the same resolution and coordinate definition would be required, or the evaluation would need independent left–right consistency errors, reprojection errors, and manually annotated regions. Even when the official `aloeGT.png` is retained, its encoding scale, valid region, and geometric correspondence with the current images must first be checked before computing MAE, RMSE, or the bad-pixel ratio. Until those conditions are verified, the reference image cannot be used directly as ground truth for this experiment. The conclusion of the current report is therefore limited to the statement that it produces an interpretable relative disparity structure, rather than claiming a particular disparity accuracy.

6.5 From numerical distributions to spatial structure

The disparity range and quantiles indicate whether the numerical values lie within the matcher's working range, but they do not by themselves show where errors occur. Spatially, occlusion boundaries, weak-texture regions, and the image-side search boundaries may all appear as invalid pixels. The same invalid fraction can therefore correspond to very different spatial patterns. The results should consequently be interpreted by treating the valid-disparity mask as a spatial support map and inspecting it alongside the left-image texture, the disparity grayscale image, and the relative-depth map. This separates a region with insufficient evidence over its whole area from a small group of rejected pixels near an object boundary, instead of summarizing all failure modes with one global percentage.

Similarly, a region with larger disparity can generally be interpreted as relatively closer, but an isolated high-disparity patch may also be an incorrect match caused by repeated texture. SGBM smoothness uses neighbourhood continuity to supplement local evidence, so a continuous result may reflect a real surface or the algorithm's prior. A local disparity change should be interpreted as a front-to-back change only when its spatial structure, left–right consistency, and available reference evidence support that interpretation together.

6.6 Different effects of a single-pixel perturbation in far and near regions

As shown by Equation (5), depth inversion is nonlinear. For the same absolute disparity error Δd, the relative error magnitude is larger in a low-disparity region, which generally corresponds to a farther scene region. Consequently, even a matching deviation on the order of one pixel in a

distant weak-texture region may cause a more noticeable perturbation in relative-depth ordering. A near region has larger disparity, so its front-to-back ordering is often easier to separate, but occlusion and boundary mixing can make local matching more difficult. The formula alone therefore does not guarantee that far regions are numerically more stable or that near regions are always more accurate. Evaluation should consider disparity magnitude, texture conditions, and boundary location together.

This nonlinearity is also why the experiment retains floating-point disparity instead of saving only an 8-bit grayscale image. The display image is stretched between the first and 99th percentiles and is intended primarily for structural inspection; it compresses extreme values and loses the numerical meaning of invalid pixels. Any analysis of disparity perturbations, depth changes, or quantiles must return to `disparity_raw.npy` and `depth_proxy.npy`, rather than attempting to infer the original pixel displacement from display grayscale values.

6.7 Low-cost improvements

Within the scope of the course task, the following low-cost improvements are possible:

1. Capture a sequence of left–right checkerboard images with known square size for the same stereo system, and estimate the intrinsics, distortion, extrinsics, and Q matrix;
2. Use the calibration parameters to undistort and stereo-rectify the images before running the same SGBM and array-saving pipeline;
3. While retaining the original `npy` arrays, compare a small, pre-specified set of `numDisparities`, `blockSize`, and speckle parameter settings, saving each parameter set and result separately;
4. Add left–right consistency and confidence statistics to the disparity output in order to distinguish "no match" from "low-confidence match".

These improvements do not have the same priority. If the goal is physical measurement, calibration and rectification for the same camera system should come first. If the goal is only to compare the spatial structure of the matcher, pre-specified parameter comparisons and consistency statistics should come first. In either case, inputs, resolution, search range, and output files should remain one-to-one, so that disparity arrays from different experimental conditions are not mixed into a single conclusion. These are improvement suggestions; this submission does not claim to have completed the additional calibration or comparison experiments.

7 Conclusion

7.1 Contribution

Using the official OpenCV aloe stereo image pair, this experiment establishes a complete pipeline from data loading, dimension checking, grayscale conversion and synchronized downsampling, through StereoSGBM matching, to floating-point disparity, the valid-disparity mask, a relative-depth proxy, and result archiving. The report also retains the Mermaid flowchart source, so that the processing steps can be checked both from the diagram and from the program entry point.

7.2 Evidence

Statistics from the real arrays show that 261378 of the 355755 pixels satisfy the valid-disparity condition, corresponding to 73.4713%. The valid disparity ranges from 16.000 to 105.875 px,

and the relative-depth proxy ranges from 0.0094451 to 0.0625. Direct array verification gives a maximum error of 0.0 for $1/d$ in the saved relative-depth array, showing that the saved proxy is consistent with the formula used in the report. The result images reveal the main structures of the leaves, flowerpot, and background, supporting the experimental conclusion that disparity represents relative front-to-back structure.

7.3 Implications and limitations

The experiment completes OpenCV stereo matching and relative-depth estimation on an arbitrary scale, and demonstrates the distinction between parameter settings, invalid-region filtering, and raw arrays versus display images. Because the aloe stereo pair has no explicitly paired focal length f, baseline B, or $\mathbf{Q}$ matrix, this experiment cannot convert `depth_proxy` into metres or centimetres and does not output a metric 3D point cloud. Occlusion, insufficient texture, repeated texture, and the image-boundary search range further reduce the valid-pixel fraction.

For metric depth measurement, the same stereo system should first be calibrated and stereo-rectified. The matching, array-saving, and statistical-verification procedures from this experiment can then be reused. The reported scope and limitations are based on the real run; unperformed calibration and comparison steps are not presented as experimental facts.